

AIMS System - Detailed Class Design Documentation

1. Design Class: Order

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	orderId	Int	"Pending"	Unique identifier for the order
2	status	String	empty list	Current status: "pending," "approved," "rejected," "canceled," etc.
3	orderItems	List<OrderItem>	empty list	Collection of items (products + quantity) in this order
4	rushOrderItems	List<OrderItem>	empty list	Collection of items eligible for rush delivery
5	regularOrderItems	List<OrderItem>	empty list	Collection of items for regular delivery
6	createdDate	DateTime	current time	Timestamp when the order is first created
7	subtotal	float	0.0	Sum of all product prices excluding VAT
8	vatAmount	float	0.0	Calculated VAT amount (10% of subtotal)
9	regularDeliveryFee	float	0.0	Shipping fee for regular delivery items
10	rushDeliveryFee	float	0.0	Shipping fee for rush delivery items
11	totalAmount	float	0.0	Total amount including VAT and all delivery fees
12	deliveryInfo	DeliveryInfo	null	Delivery information associated with this order
13	paymentTransactionId	int	null	Reference to payment transaction
14	VAT_RATE	float	0.1	Value-added tax rate (constant 10%)

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	Order()	void	Constructor for creating a new Order object
2	updateToReject()	void	Sets the order's status to "rejected"
3	updateToApprove()	void	Sets the order's status to "approved"
4	displayApproveStatus()	void	Shows or logs that the order has been approved
5	getOrderDetails	Order	Retrieves/shows the order's item list, price, etc.
6	updateStatus()	void	Updates this order's status based on input
7	save()	void	Persists this order in storage
8	calculateSubtotal	float	Calculates the sum of all product prices excluding VAT
9	calculateVAT	float	Calculates the VAT amount based on subtotal
10	calculateRegularDeliveryFee	float	Calculates shipping fee for regular delivery items
11	calculateRushDeliveryFee	float	Calculates shipping fee for rush delivery items
12	calculateTotalAmount	float	Calculates the total amount including VAT and all delivery fees
13	isEligibleForFreeShipping	boolean	Checks if order is eligible for free regular shipping
14	createOrderListScreen	float	Creates a screen listing multiple orders (for manager UI)
15	getOrderById	Order	Retrieves an order by its ID

Parameter:

- **updateStatus(newStatus: String): void**
 - **newStatus:** The updated status ("pending", "approved", "rejected", "canceled", etc.).
- **getOrderById(orderId: int): Order**
 - **orderId:** The identifier of the order to retrieve.
 - **Return:** The Order object if found; otherwise might throw an exception.
- **calculateSubtotal(orderItems: List<OrderItem>): float**
 - **orderItems:** The list of ordered items to sum up.
 - **Return:** The subtotal of all items excluding VAT.
- **calculateVAT(subtotal: float): float**
 - **subtotal:** The total cost of items before VAT.
 - **Return:** subtotal * VAT_RATE.
- **isEligibleForFreeShipping(): boolean**
 - No parameters. Uses internal order data (subtotal, location, etc.).
 - **Return:** true if the order meets free shipping criteria, otherwise false.

- Exception:
- **OrderNotFoundException**
 - Thrown if no order exists with the specified orderId when calling methods like getOrderById().
 - **OrderAlreadyProcessedException**
 - Thrown if attempting to approve, reject, or cancel an order that has already been processed (i.e., not in "pending").

- Method:
- **Order() (constructor)**
 - **Purpose:** Initializes a new order object with default values (status = "pending", createdAt = current time).
 - **Usage:** Called when the system or user creates an order (e.g., after finalizing a cart).
 - **updateToReject()**
 - **Purpose:** Changes the order's status to "rejected".
 - **Usage:** Called by a manager or system logic if an order cannot be fulfilled (out of stock, other reasons).
 - **updateToApprove()**
 - **Purpose:** Changes the order's status to "approved".
 - **Usage:** Invoked once the manager or system confirms items are in stock and the order is valid.
 - **save()**
 - **Purpose:** Persists this order's current data (status, items, totalAmount, etc.) to a database or file.
 - **Usage:** Called whenever the order state is updated (e.g., after approving or rejecting).
 - **calculateSubtotal(), calculateVAT(), calculateRegularDeliveryFee(), calculateRushDeliveryFee(), calculateTotalAmount()**
 - **Purpose:** Each method breaks down the cost calculation to keep the code organized.
 - **Usage:** Typically chained together after the user finalizes or updates the order's items or delivery method.
 - **isEligibleForFreeShipping()**
 - Purpose: Checks rules (like "subtotal > 100,000 VND" or other promotional logic) to see if shipping can be waived.
 - Usage: Called during fee calculation to adjust regularDeliveryFee if free shipping applies.

- State
- If status = "approved" or "rejected" → The order is considered processed and cannot be changed to another status without special rules.
 - If orderItems changes → Recalculate subtotal, vatAmount, rushDeliveryFee, regularDeliveryFee, totalAmount.
 - If free shipping is applicable → regularDeliveryFee might be set to 0 or partial discount.
 - If paymentTransactionId is set → A corresponding payment has been recorded for this order.

2. Design Class: OrderController

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	currentOrder	Order	null	Currently selected order
2	orderList	List<Order>	empty	A list of orders loaded for viewing or processing
3	alreadyProcessed	boolean	false	Indicates if an order has already been processed or canceled

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	approveOrder	void	Approves a pending order
2	rejectOrder	void	Rejects a pending order with reason
3	viewOrderList	void	Retrieves or displays a list of pending orders.
4	getOrderDetails	void	Gets details for a specific order
5	checkAvailableInventory	boolean	Checks if there's enough inventory for an order
6	processCancel	boolean	Processes the cancellation of an order
7	createOrder	Order	Creates a new order in the system
8	refundConfirm	void	Initiates or confirms a refund operation via VNPay.
9	selectCancel	boolean	Handles user requests to cancel the order.
10	checkAlreadyProcessed	boolean	Checks if the current order is already processed (approved/rejected).
11	getOrder	Order	Returns the current Order object in the controller

- Parameter.
- checkAlreadyProcessed(orderId: String): boolean

- orderId: The ID of the order to check
 - Return: true if the order is already approved or rejected, otherwise false.
- selectCancel(orderId: String): boolean
 - orderId: The order to attempt to cancel.
 - Return: true if cancellation is selected, false otherwise.
- refundConfirm(orderId: String): void
 - orderId: The order to be refunded.
 - processCancel(orderId: String): boolean
- ApproveOrder(orderId: String): void
 - orderId: The ID of the order to approve
- RejectOrder(orderId: String): void
 - orderId: The ID of the order to reject
- processCancel(orderId: String): boolean
 - orderId: The ID of the order to finalize cancellation for.
 - Return: true if successfully canceled, otherwise false.
- GetOrderDetail(orderId: String)
 - orderId: The ID of the order whose details are to be fetched.
 - Return: detail information

Exception

- OrderNotFoundException: If the specified orderId does not exist.
- OrderAlreadyApprovedException: If trying to approve/reject/cancel an order that is already processed.

Method

- approveOrder(orderId)
 - Purpose: Set order status to "approved" and decrement inventory.
 - Usage: Called by product manager or automated system upon reviewing an order.
- rejectOrder(orderId)
 - Purpose: Set order status to "rejected". Possibly triggers a refund if payment was collected.
- checkAlreadyProcessed(orderId)
 - Purpose: Quickly verify whether an order is no longer in "pending" state.
- selectCancel(orderId) and processCancel(orderId)
 - Purpose: used in two-step cancellation processes.

State

- If orderStatus is approved or rejected → The order is processed, thus cannot be changed.
- If product is insufficient → approveOrder might either reject the order or throw an exception.